



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS
2018

HSC Task #1

Mathematics Extension II

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** question in a separate answer booklet.

Total Marks - 65

Section I	Pages	2-4
5 marks		

- Attempt questions 1-5
- Allow about 10 minutes for this section

Section II	Pages	5-10
60 Marks		

- Attempt Questions 6-9
- Allow about 1 hour and 20 minutes for this section

Examiner: **A. Wu**

1 Section I - Multiple Choice

5 Marks

Attempt questions 1-5

Allow about 10 minutes for this section

Use the multiple choice answer sheet for questions 1-5

1. What is the correct factorisation of the $x^3 + i$?

A. $(x - i)(-1 + ix + x^2)$

B. $(x + i)(1 + ix - x^2)$

C. $(x + i)(1 + x + x^2)$

D. $(x - i)(1 + x + ix^2)$

2. The polynomial $P(x) = x^3 - ax + 2$ has a positive double zero. What is a ?

A. $-3\sqrt[3]{3}$

B. -3

C. $3\sqrt[3]{3}$

D. 3

3. Let ω be a complex cube root of unity (i.e. $\Im(\omega) \neq 0$). What is the sum of the roots of the following polynomial?

$$\sum_{k=1}^{2018} (x - \omega^k)^{2018}$$

A. 0

B. -2018

C. 2018ω

D. $2018(\omega^2 - 1)$

4. Suppose z, w are complex numbers. Select the inequality that is not always true.

A. $|z + w| \leq |z| + |w|$

B. $||z| - |w|| \leq |z - w|$

C. $|z| - |w| \leq |z - w|$

D. $||z| + |w|| \geq |z - w| + |w|$

5. If $\arg\left(\frac{z - \sqrt{3} - 2i}{z - \sqrt{3}}\right) = \frac{\pi}{2}$, what is the maximum value of $\arg(z)$?

A. $\frac{\pi}{2}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{3}$

D. $\tan^{-1} \frac{1}{2}$

Bonus Question: What is the value of $1+2+3+4+\dots$

A. ∞

B. 1

C. 0

D. $-\frac{1}{12}$

2 Section II - Extended Response

65 Marks

Attempt questions 6-9

Allow about 1 hour and 35 minutes for this section

Answer each question in a NEW writing booklet. Extra pages are available

In Questions 6–9, your responses **must include** relevant mathematical reasoning and/or calculations.

6. Question 6 (15 marks)

Start a NEW writing booklet.

(a) Let $z = 1 + i$, $w = \sqrt{3} + i$.

i. 1 Rewrite w in mod-arg form

ii. 1 Find w^{2018} in cartesian form

iii. 1 Rewrite $\frac{z}{w}$ in cartesian form

iv. 2 Deduce that

$$\cos \frac{\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

(b) Sketch the following loci on the Argand Diagram

i. 2 $|z - 2i| = \Im(z)$

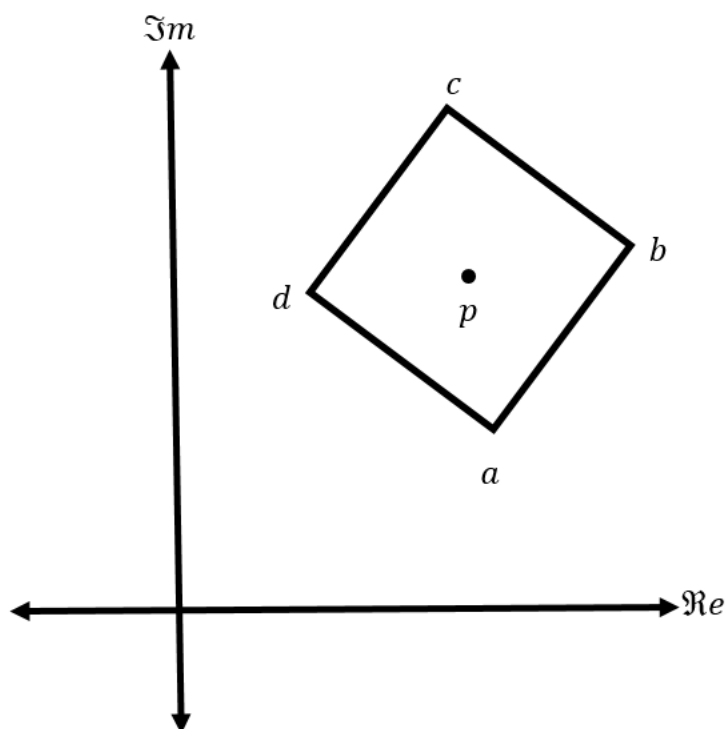
ii. 2 $\Im(z) = \tan(\arg z)$ for $0 \leq \arg z \leq \frac{\pi}{2}$

(c) i. **1** Write $\frac{1}{(a+i)(b+i)}$ with a real denominator

ii. **2** Hence, by using arguments, for $ab > 1$ and $a, b > 0$, prove that

$$\tan^{-1}\left(\frac{1}{a}\right) + \tan^{-1}\left(\frac{1}{b}\right) = \tan^{-1}\left(\frac{a+b}{ab-1}\right)$$

(d) **3**



Suppose the complex numbers a, b, c, d are the vertices of a square in anti-clockwise order. Let p be the centre of the square (midpoint and intersection of diagonals). Show that

$$p = \frac{a + d + i(b - c)}{2}$$

End of Question 6

7. Question 7 (15 marks)

Start a NEW writing booklet.

(a) Consider the polynomial equation $x^3 - \left(\frac{1 + \sqrt{5}}{2}\right)x^2 + \left(\frac{1 + \sqrt{5}}{2}\right)x - 1 = 0$. It is known that $x = 1$ is a root.

i. 2 Prove that at least one root is non-real.

ii. 2 Deduce that the other two roots lie on the unit circle

(b) 2 Let the polynomial $P(x) = x^4 - x^3 + 5x^2 - 2x + 7$ have zeroes $\alpha, \beta, \gamma, \delta$. Evaluate:

$$(\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1)$$

Hint: Do NOT expand

(c) At the start of the year, n scholar sportsmen have the potential to achieve the Joseph Coates award by the end of the year.
Depending on procrastination habits and dedication to GPS sports, any number of students k , where $0 \leq k \leq n$, $k \in \mathbb{Z}$ may receive the award.

Assume that all n students are equally likely to achieve or to not achieve the award and that one student's achievement is independent of another's.

i. 1 Show that the probability of k students winning the Joseph Coates Award is

$$\binom{n}{k} \frac{1}{2^n}$$

ii. 1 Deduce that

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

- (d) i. **2** Let z, w be complex numbers with $\theta = \arg w > \arg z = \phi$. Prove that:

$$|z - w|^2 = |z|^2 + |w|^2 - 2|z||w| \cos(\theta - \phi)$$

- ii. **1** If z, w lie on the unit circle, show that they are separated by a distance of

$$2 \sin\left(\frac{\theta - \phi}{2}\right)$$

- (e) **4** Using mathematical induction, prove the General Leibniz rule (below) for positive integers n , where $f^{(n)}(x)$ represents the n^{th} derivative of $f(x)$.

You are given that $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ for positive integers $n \geq k$.

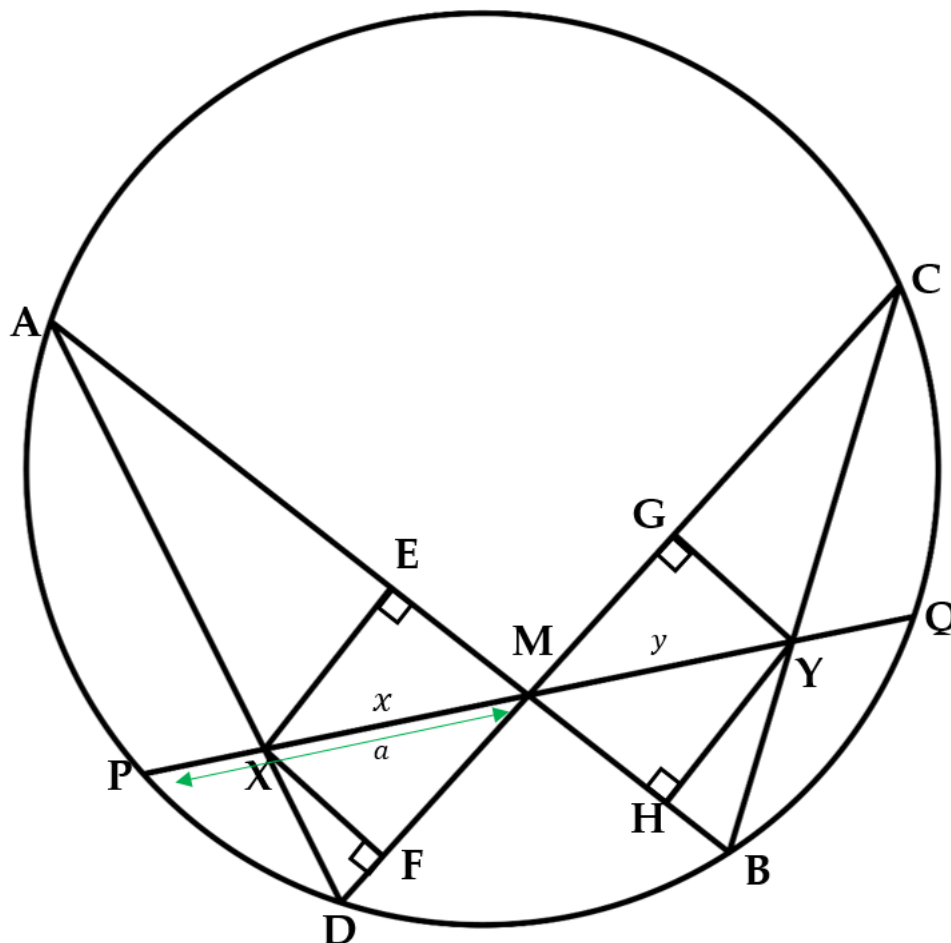
$$\frac{d^n}{dx^n} [f(x)g(x)] = \sum_{k=0}^n \binom{n}{k} f^{(n-k)}(x)g^{(k)}(x)$$

End of Question 7

8. Question 8 (15 marks)

Start a NEW writing booklet.

- (a) *The Butterfly Theorem*: Let M be the midpoint of a chord PQ of a circle. Two more chords AB and CD are drawn through M . AD and BC intersect the chord PQ at points X, Y respectively. Let E, F be the feet of the perpendiculars drawn from X to AM and DM respectively. Similarly, let G, H be the feet of the perpendiculars drawn from Y to CM and BM respectively.



Let $PM = a = QM$, $MX = x$, $MY = y$.

- i. 1 Prove that $\frac{x}{y} = \frac{EX}{HY}$
- ii. 1 Prove that $\frac{EX}{GY} = \frac{AX}{CY}$

- iii. **[3]** Hence or otherwise, show that

$$\frac{x^2}{y^2} = \frac{AX \cdot XD}{CY \cdot YB}$$

- iv. **[2]** Show that

$$\frac{AX \cdot XD}{CY \cdot YB} = \frac{a^2 - x^2}{a^2 - y^2}$$

- v. **[1]** Deduce that $x = y$

(b) *Chebyshev Polynomials:*

- i. **[2]** Let $T_n(x) = \cos(n \cos^{-1} x)$, $U_n(x) = \sin(n \cos^{-1} x)$ where $|x| \leq 1$.

Using DeMoivre's Theorem, prove that

$$T_n(x) + i \cdot U_n(x) = \left(x + i\sqrt{1 - x^2}\right)^n$$

- ii. **[2]** Hence, prove that:

$$T_n(x) = \frac{1}{2} \left[\left(x + \sqrt{x^2 - 1}\right)^n + \left(x - \sqrt{x^2 - 1}\right)^n \right]$$

- iii. **[3]** Consider the series $S = \sum_{n=0}^{\infty} t^n (\cos \theta + i \sin \theta)^n$, where $-1 < t < 1$ and $\theta = \cos^{-1} x$.

Prove that:

$$\sum_{n=0}^{\infty} T_n(x) t^n = \frac{1 - tx}{1 - 2tx + t^2}$$

End of Question 8

9. Question 9 (15 marks)

Start a NEW writing booklet.

- (a) At lunchtime, 5 senior students and 6 junior students decide to line up at the canteen to purchase Chicken Pides. The canteen has three lines for students to purchase food; 2 (fixed) lines for juniors and 1 for seniors. The seniors, having much respect for the juniors, decide that they will line up wherever they want. The juniors, however, stick to their own lines.

- i. **1** In how many ways can the 5 seniors be distributed amongst the 3 lines?
- ii. **3** In how many ways can the 5 seniors and 6 juniors be distributed among the 3 lines?

- (b) Consider the general polynomial cubic equation $P(x) = ax^3 + bx^2 + cx + d = 0$ for $a \neq 0$ and real constants a, b, c, d . $P(x)$ has roots α, β, γ

- i. **2** Show that the following polynomial has roots $\alpha + \frac{b}{3a}, \beta + \frac{b}{3a}, \gamma + \frac{b}{3a}$

$$ay^3 + \left(c - \frac{b^2}{3a}\right)y + \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a}\right) = 0 \quad (1)$$

- ii. **1** Let $p = \frac{1}{a} \left(c - \frac{b^2}{3a}\right), q = \frac{1}{a} \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a}\right)$.

Using the substitution $y = s + t$, where s, t are real variables, show that equation (1) is equivalent to equation (2).

$$s^3 + t^3 + (3st + p)(s + t) + q = 0 \quad (2)$$

- iii. **1** Suppose that $3st + p = 0$. Show that

$$s^3 + t^3 = -q$$

$$s^3 t^3 = -\frac{p^3}{27}$$

- iv. **[2]** Assume without loss of generality that $s^3 > t^3$. Prove that:

$$s^3 = -\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}, \quad t^3 = -\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}$$

- v. **[2]** Show that the possible solutions of s are:

$$s = \omega^k \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } k = 0, 1, 2$$

where ω is the complex cube root of unity with positive imaginary part.

- vi. **[3]** Show that the roots of equation $P(x) = 0$ are

$$x = -\frac{b}{3a} + \omega^k \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } k = 0, 1, 2$$

where ω is the complex cube root of unity with positive imaginary part.

End of Examination



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Examiner: **A. Wu**

Note: These are quick solutions. You may need to provide more working in an exam

1 Section I - Multiple Choice

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Use the multiple choice answer sheet for questions 1-5

1. What is the correct factorisation of the $x^3 + i$?

A. $(x - i)(-1 + ix + x^2)$

B. $(x + i)(1 + ix - x^2)$

C. $(x + i)(1 + x + x^2)$

D. $(x - i)(1 + x + ix^2)$

Explanation: Write $x^3 + i$ as $x^3 - i^3$ and use the difference of cubes factorisation.

2. The polynomial $P(x) = x^3 - ax + 2$ has a positive double zero. What is a ?

A. $-3\sqrt[3]{3}$

B. -3

C. $3\sqrt[3]{3}$

D. 3

Explanation: $P'(x) = 3x^2 - a = 0$. But the double root is positive, so take $x = \sqrt{\frac{a}{3}}$ and substitute back into $P(x)$ to give $a = 3$.

3. Let ω be a complex cube root of unity (i.e. $\Im(\omega) \neq 0$). What is the sum of the roots of the following polynomial?

$$\sum_{k=1}^{2018} (x - \omega^k)^{2018}$$

A. 0

B. -2018

C. 2018ω

D. $2018(\omega^2 - 1)$

Explanation: let S be the sum of the roots of the polynomial. By the Binomial theorem (or Pascal's triangle), we have:

$$\begin{aligned} S &= 2018(\omega + \omega^2 + \omega^3 + \dots + \omega^{2018}) \\ &= 2018(-1 + 1 + \omega + \omega^2 + \omega^3(1 + \omega + \omega^2) + \dots + \omega^{2016}(1 + \omega + \omega^2)) \\ &= -2018 \quad (\omega^3 = 1, 1 + \omega + \omega^2 = 0) \text{ since } \omega \text{ is a complex cube root of unity} \end{aligned}$$

4. Suppose z, w are complex numbers. Select the inequality that is not always true.

A. $|z + w| \leq |z| + |w|$

B. $||z| - |w|| \leq |z - w|$

C. $|z| - |w| \leq |z - w|$

D. $||z| + |w|| \geq |z - w| + |w|$

Explanation: All inequalities from A to C are true for all vectors (this can be proved using geometry). D is easily seen as not true if we sub in $z = 0$.

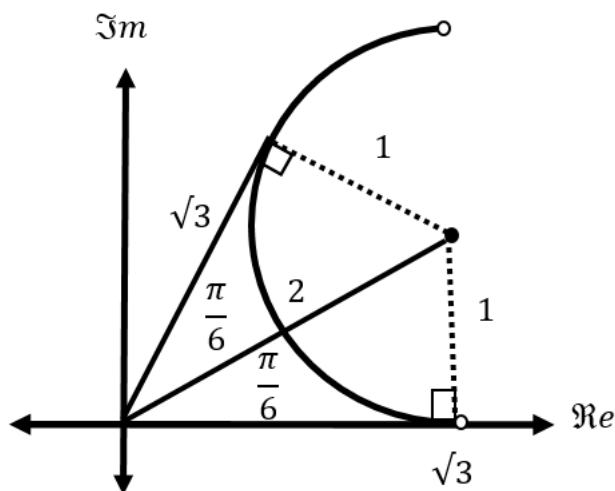
5. If $\arg\left(\frac{z - \sqrt{3} - 2i}{z - \sqrt{3}}\right) = \frac{\pi}{2}$, what is the maximum value of $\arg(z)$?

A. $\frac{\pi}{2}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{3}$

D. $\tan^{-1} \frac{1}{2}$



Explanation: The Locus is a semicircle radius 1, center $\sqrt{3} + i$. The maximum value occurs when a ray from the origin is tangent to the circle (and makes a right angle at the point of contact by circle geometry). Using trigonometry we then obtain the maximum value of $\arg z$ as $\frac{\pi}{3}$.

Bonus Question: What is the value of $1+2+3+4+\dots$

A. ∞

B. 1

C. 0

D. $-\frac{1}{12}$

Explanation: No comment.

2 Section II - Extended Response

65 Marks

Attempt questions 6-9

Allow about 1 hour and 35 minutes for this section

Answer each question in a NEW writing booklet. Extra pages are available

In Questions 6–9, your responses **must include** relevant mathematical reasoning and/or calculations.

6. Question 6 (15 marks)

Start a NEW writing booklet.

(a) Let $z = 1 + i$, $w = \sqrt{3} + i$.

i. 1 Rewrite w in mod-arg form

$$2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

ii. 1 Find w^{2018} in cartesian form

$$2^{2017}(1 + i\sqrt{3})$$

iii. 1 Rewrite $\frac{z}{w}$ in cartesian form

$$\frac{\sqrt{3} + 1}{4} + i \frac{\sqrt{3} - 1}{4}$$

iv. 2 Deduce that

$$\cos \frac{\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

Using mod-arg forms we have:

$$\frac{z}{w} = \frac{\sqrt{2} \operatorname{cis} \frac{\pi}{4}}{2 \operatorname{cis} \frac{\pi}{6}} = \frac{1}{\sqrt{2}} \operatorname{cis} \frac{\pi}{12}$$

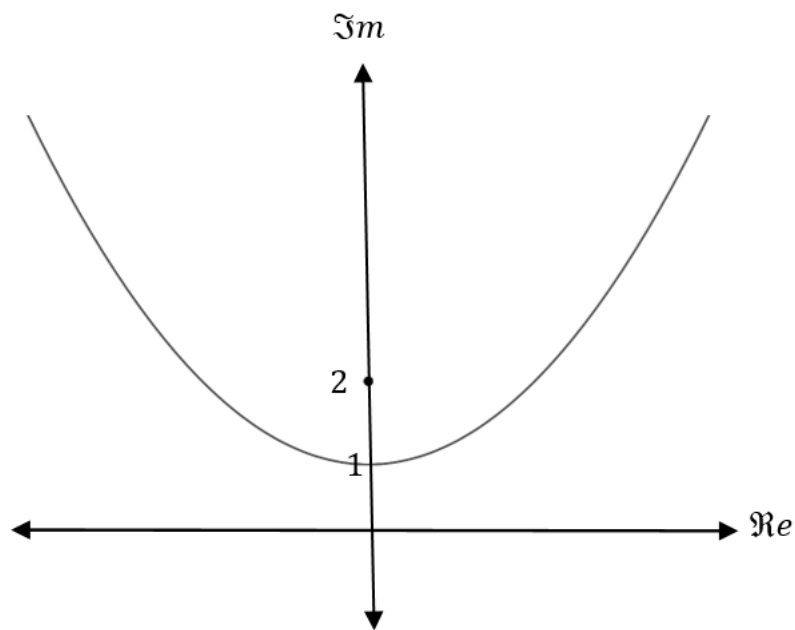
Equating real parts with iii) gives us

$$\begin{aligned} \frac{1}{\sqrt{2}} \cos \frac{\pi}{12} &= \frac{\sqrt{3} + 1}{4} \\ \cos \frac{\pi}{12} &= \frac{\sqrt{2} + \sqrt{6}}{4} \end{aligned}$$

(b) Sketch the following loci on the Argand Diagram

i. 2 $|z - 2i| = \Im(z)$

$\Im(z)$ is the real line since the imaginary part of a complex number is real. So this locus is all points such that z is equidistant from the real line and the point $2i$, that is, a parabola with focus $2i$ and directrix $\Im(z) = 0$



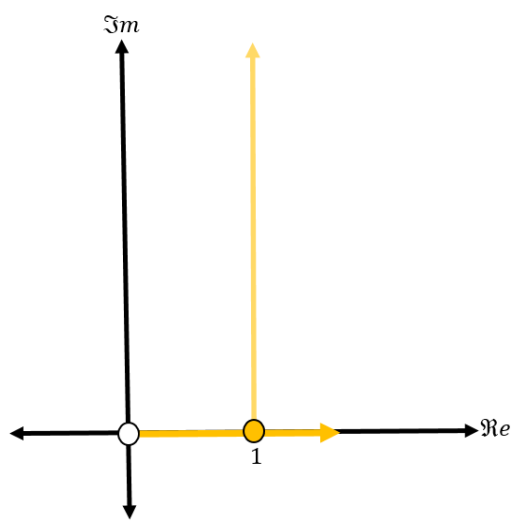
- ii. **2** $\Im m(z) = \tan(\arg z)$ for $0 \leq \arg z \leq \frac{\pi}{2}$

Since $0 \leq \arg z \leq \frac{\pi}{2}$, z lies in the first quadrant. Hence if $z = x + iy$, then the locus becomes

$$y = \frac{y}{x}$$

$$y \left(1 - \frac{1}{x} \right) = 0$$

which gives $y = 0$ or $x = 1$. But recall that z is in the first quadrant, so take only the halves of these lines that lie in the first quadrant. Open circle at origin because $\arg z$ is undefined.



- (c) i. **1** Write $\frac{1}{(a+i)(b+i)}$ with a real denominator

$$\begin{aligned}\frac{1}{(a+i)(b+i)} &= \frac{(a-i)(b-i)}{(a+i)(a-i)(b-i)(b+i)} \\ &= \frac{ab-1-i(a+b)}{(a^2+1)(b^2+1)}\end{aligned}$$

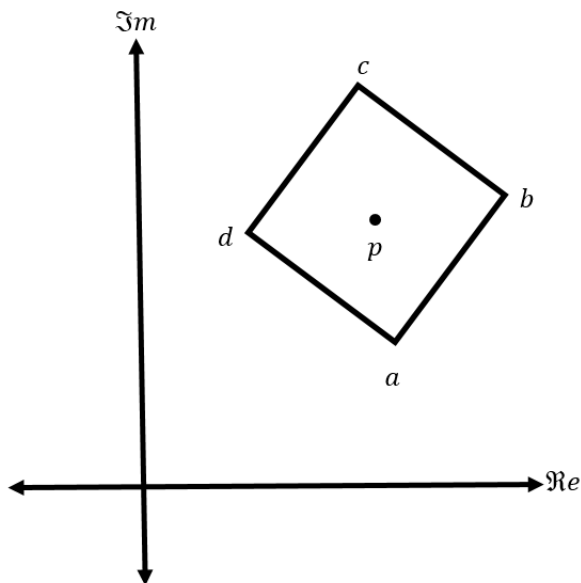
- ii. **2** Hence, by using arguments, prove that $\tan^{-1}\left(\frac{1}{a}\right) + \tan^{-1}\left(\frac{1}{b}\right) = \tan^{-1}\left(\frac{a+b}{ab-1}\right)$

$$\arg\left(\frac{1}{(a+i)(b+i)}\right) = \arg\left(\frac{ab-1-i(a+b)}{(a^2+1)(b^2+1)}\right)$$

$$\arg 1 - \arg(a+i) - \arg(b+i) = \tan^{-1}\left(-\frac{a+b}{ab-1}\right) \quad \text{since number is in first quadrant}$$

$$\tan^{-1}\left(\frac{1}{a}\right) + \tan^{-1}\left(\frac{1}{b}\right) = \tan^{-1}\left(\frac{a+b}{ab-1}\right) \quad \text{since } \tan^{-1} \text{ is odd}$$

- (d) **3**



Suppose the complex numbers a, b, c, d are the vertices of a square in anti-clockwise order. Let p be the centre of the square (where its diagonals intersect). Show that

$$p = \frac{a+d+i(b-c)}{2}$$

Let the complex numbers a, b, c, d be represented by points A, B, C, D respectively.

$$\overrightarrow{AP} + \overrightarrow{PB} = \overrightarrow{AB}$$

$$p - a + i(p - a) = d - a \quad \text{diagonals of square are perpendicular}$$

$$p = \frac{d + ia}{1 + i}$$

$$= \frac{a + d + i(a - d)}{2} \quad \text{multiplying by } \frac{1 - i}{1 - i}$$

$$= \frac{a + d + i(b - c)}{2} \quad \text{opposite sides of a square are equal}$$

7. Question 7 (15 marks)

Start a NEW writing booklet.

(a) Consider the polynomial equation $x^3 - \left(\frac{1 + \sqrt{5}}{2}\right)x^2 + \left(\frac{1 + \sqrt{5}}{2}\right)x - 1 = 0$. It is known that $x = 1$ is a root.

i. **2** Prove that at least one the roots is non-real.

Let the other two roots be a, b

$$\begin{aligned} 1 + a^2 + b^2 &= (1 + a + b)^2 - 2(a + b + ab) \\ a^2 + b^2 &= -1 + \left(\frac{1 + \sqrt{5}}{2}\right)^2 - 2\left(\frac{1 + \sqrt{5}}{2}\right) \\ &= -\frac{1 + \sqrt{5}}{2} \\ &< 0 \end{aligned}$$

As the sum of their squares is negative, at least one of roots a, b must be non-real.

ii. **2** Deduce that the other two roots lie on the unit circle

Let the roots be $z, \bar{z}$ (by the conjugate root theorem, real polynomials must have complex roots in conjugate pairs).

By the product of roots we have $z\bar{z} = |z|^2 = 1$ so $|z| = |\bar{z}| = 1$ and hence the roots lie on the unit circle.

(b) **2** Let the polynomial $P(x) = x^4 - x^3 + 5x^2 - 2x + 7$ have zeroes $\alpha, \beta, \gamma, \delta$. Evaluate:

$$(\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1)$$

Hint: Do NOT expand

$$\begin{aligned}
P(x) &= (x - \alpha)(x - \beta)(x - \gamma)(x - \delta) \\
P(-1) &= (\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1) \\
(\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1) &= (-1)^4 - (-1)^3 + 5(-1)^2 - 2(-1) + 7 \\
&= 16
\end{aligned}$$

- (c) At the start of the year, n scholar sportsmen have the potential to achieve the Joseph Coates award by the end of the year.

Depending students procrastination habits and dedication to GPS sports, any number of students k , where $0 \leq k \leq n$, $k \in \mathbb{Z}$, may receive the award. Assume that all n students are equally likely to achieve or to not achieve the award and that one student's achievement is independent of another's.

- i. **1** Show that the probability of k students winning the Joseph Coates Award is

$$\binom{n}{k} \frac{1}{2^n}$$

There are $\binom{n}{k}$ ways to choose k students from n and the probability that k students achieve the award is $\left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k}$. Multiplying these gives $\binom{n}{k} \frac{1}{2^n}$.

- ii. **1** Deduce that

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\begin{aligned}
\sum_{k=0}^n \binom{n}{k} \frac{1}{2^n} &= 1 \quad \text{sum probabilities of all possible outcomes} \\
\therefore \sum_{k=0}^n \binom{n}{k} &= 2^n
\end{aligned}$$

- (d) i. **2** Let z, w be complex numbers with $\theta = \arg w > \arg z = \phi$. Prove that:

$$|z - w|^2 = |z|^2 + |w|^2 - 2|z||w| \cos(\theta - \phi)$$

$$\begin{aligned}
|z - w|^2 &= (z - w)(\bar{z} - \bar{w}) \\
&= z\bar{z} + w\bar{w} - 2(w\bar{z} + \overline{w\bar{z}}) \\
&= |z|^2 + |w|^2 - 2\Re(|w| \operatorname{cis} \theta |z| \operatorname{cis}(-\phi)) \\
&= |z|^2 + |w|^2 - 2\Re(|w||z| \operatorname{cis}(\theta - \phi)) \\
&= |z|^2 + |w|^2 - 2|z||w| \cos(\theta - \phi)
\end{aligned}$$

- ii. **1** If z, w lie on the unit circle, show that they are separated by a distance of $2 \sin \left(\frac{\theta - \phi}{2} \right)$ By part i we have

$$\begin{aligned} |\operatorname{cis} \theta - \operatorname{cis} \phi|^2 &= 2 - 2 \cos(\theta - \phi) \\ &= 4 \sin^2 \left(\frac{\theta - \phi}{2} \right) \quad \text{double angle formula} \\ |\operatorname{cis} \theta - \operatorname{cis} \phi| &= 2 \sin \left(\frac{\theta - \phi}{2} \right) \quad \text{modulus is positive} \end{aligned}$$

- (e) **4** Using mathematical induction, prove the General Leibniz rule (below) for positive integers n , where $f^{(n)}(x)$ represents the n^{th} derivative of the $f(x)$.

You are given that $\binom{n}{k-1} + \binom{n}{k} = \binom{n}{k+1}$ for positive integers $n \geq k$.

$$\frac{d^n}{dx^n} [f(x)g(x)] = \sum_{k=0}^n \binom{n}{k} f^{(n-k)}(x)g^{(k)}(x)$$

I: Test $n = 1$.

$$\begin{aligned} LHS &= \frac{d}{dx}(f(x)g(x)) \\ &= f^{(1)}(x)g(x) + f(x)g^{(1)}(x) \quad \text{product rule} \\ &= \sum_{k=0}^1 \binom{1}{k} f^{(1-k)}(x)g^{(k)}(x) \quad \binom{1}{0} = \binom{1}{1} = 1 \\ &= RHS \end{aligned}$$

II: Assume true for $n = r$, $r \in \mathbb{Z}^+$. Prove true for $n = r + 1$.

$$\text{i.e. } \frac{d^{r+1}}{dx^{r+1}} [f(x)g(x)] = \sum_{k=0}^r \binom{r+1}{k} f^{(r+1-k)}(x)g^{(k)}(x).$$

$$\begin{aligned} LHS &= \frac{d}{dx} \left(\frac{d^r}{dx^r} (f(x)g(x)) \right) \\ &= \frac{d}{dx} \left(\sum_{k=0}^r \binom{r}{k} f^{(r-k)}(x)g^{(k)}(x) \right) \\ &= \sum_{k=0}^r \binom{r}{k} f^{(r-k+1)}(x)g^{(k)}(x) + \sum_{k=0}^r \binom{r}{k} f^{(r-k)}(x)g^{(k+1)}(x) \quad \text{product rule} \\ &= \sum_{k=0}^r \binom{r}{k} f^{(r+1-k)}(x)g^{(k)}(x) + \sum_{k=1}^{r+1} \binom{r}{k-1} f^{(r+1-k)}(x)g^{(k)}(x) \\ &= \binom{r}{0} f^{(r+1)}(x)g(x) + \binom{r}{r} f(x)g^{(r+1)}(x) + \sum_{k=0}^r \binom{r+1}{k} f^{(r+1-k)}(x)g^{(k)}(x) \\ &\quad \text{since } \binom{r}{k-1} + \binom{r}{k} = \binom{r+1}{k} \\ &= \binom{r+1}{0} f^{(r+1)}(x)g(x) + \binom{r+1}{r+1} f(x)g^{(r+1)}(x) + \sum_{k=0}^r \binom{r+1}{k} f^{(r+1-k)}(x)g^{(k)}(x) \\ &= \sum_{k=0}^{r+1} \binom{r+1}{k} f^{(r+1-k)}(x)g^{(k)}(x) \\ &= RHS \end{aligned}$$

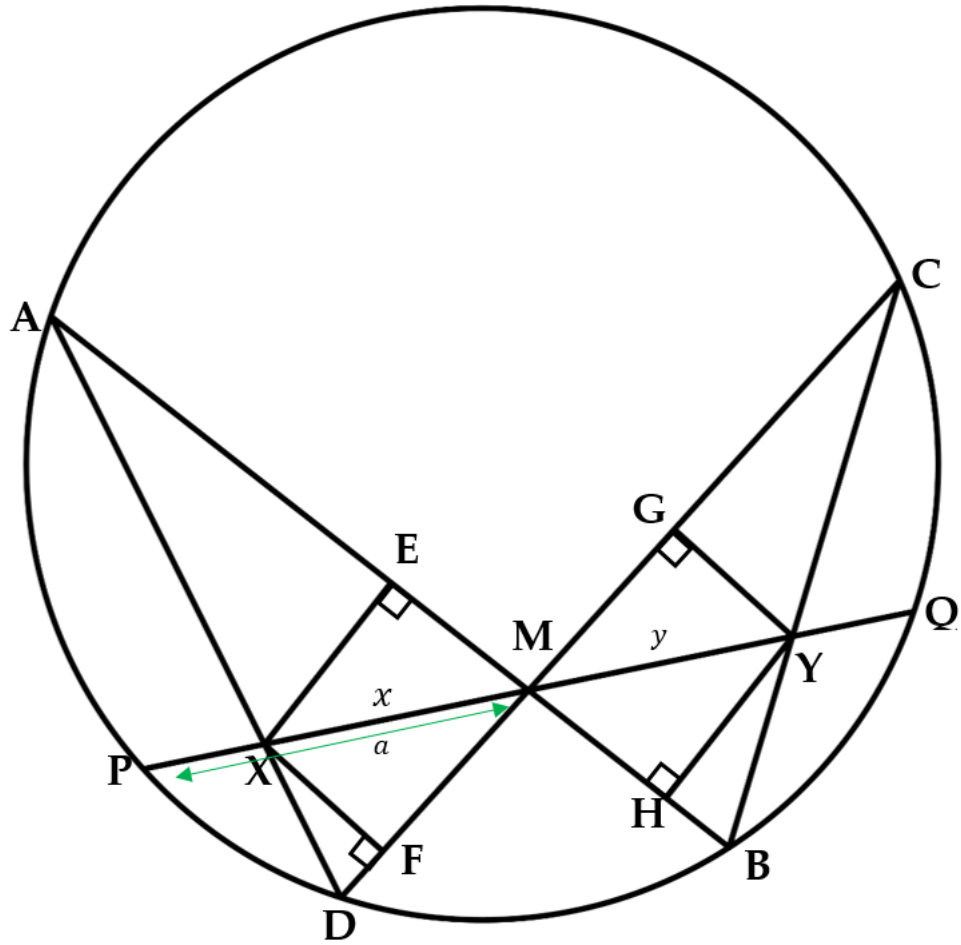
$\therefore$ true for $n = r + 1$

III: Hence the statement is true for all positive integers n

8. Question 8 (15 marks)

Start a NEW writing booklet.

- (a) *The Butterfly Theorem:* Let M be the midpoint of a chord PQ on in a circle. Two more chords AB and CD are drawn through M . AD and BC intersect the chord PQ at points X, Y respectively. Let E, F be the feet of the perpendiculars drawn from X to AM and DM respectively. Similarly, let G, H be the feet of the perpendiculars drawn from Y to CM and BM respectively.



Let $PM = a = QM$, $MX = x$, $MY = y$.

- i. **1** Prove that $\frac{x}{y} = \frac{EX}{HY}$

$$\angle XEM = \angle YHM = 90^\circ \quad \text{given}$$

$$\angle EMX = \angle HMY \quad \text{vertically opposite angles}$$

$\therefore \triangle EMX$ is similar to $\triangle HMY$

$$\therefore \frac{x}{y} = \frac{EX}{HY} \quad \text{corresponding sides of similar triangles in proportion}$$

- ii. **1** Prove that $\frac{EX}{GY} = \frac{AX}{CY}$

$$\angle AEX = \angle CGY \quad \text{given}$$

$$\angle XAE = \angle GCY \quad \text{angles in the same segment}$$

$\therefore \triangle XAE$ is similar to $\triangle GYC$

$$\therefore \frac{EX}{GY} = \frac{AX}{CY} \quad \text{corresponding sides of similar triangles in proportion}$$

iii. **3** Hence or otherwise, show that

$$\frac{x^2}{y^2} = \frac{AX \cdot XD}{CY \cdot YB}$$

Similarly to part i and ii, we have $\frac{x}{y} = \frac{FX}{GY}, \frac{FX}{HY} = \frac{DX}{BY}$. Hence

$$\begin{aligned} \frac{x^2}{y^2} &= \frac{EX}{HY} \cdot \frac{FX}{GY} \\ &= \frac{AX \cdot DX}{CY \cdot YB} \end{aligned}$$

iv. **2** Show that

$$\frac{AX \cdot XD}{CY \cdot YB} = \frac{a^2 - x^2}{a^2 - y^2}$$

$$\begin{aligned} AX \cdot XD &= PX \cdot XQ \quad \text{products of intercepts of chords} \\ &= (a + x)(a - x) \\ &= a^2 - x^2 \end{aligned}$$

Similarly, $CY \cdot YB = a^2 - y^2$, so $\frac{AX \cdot XD}{CY \cdot YB} = \frac{a^2 - x^2}{a^2 - y^2}$

v. **1** Deduce that $x = y$

By parts iii and iv

$$\frac{x^2}{y^2} = \frac{a^2 - x^2}{a^2 - y^2}$$

which rearranges to

$$a^2(x + y)(x - y) = 0$$

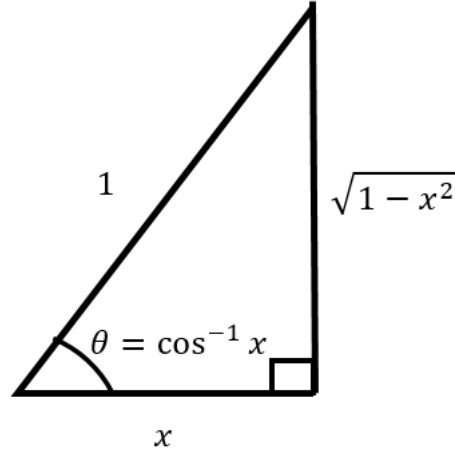
Now $x, y > 0$ so $x \neq -y$ and $a \neq 0$ so $x = y$

(b) *Chebyshev Polynomials:*

i. **2** Let $T_n(x) = \cos(n \cos^{-1} x), U_n(x) = \sin(n \cos^{-1} x)$.
Using DeMoivre's Theorem, prove that

$$T_n(x) + i \cdot U_n(x) = \left(x + i\sqrt{1 - x^2}\right)^n$$

Drawing a right triangle:



Clearly $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$.

$$\begin{aligned}
 T_n(x) + i \cdot U_n(x) &= \cos(n \cos^{-1} x) + i \sin(n \cos^{-1} x) \\
 &= (\cos(\cos^{-1} x) + i \sin(\cos^{-1} x))^n && \text{De Moivre's Theorem} \\
 &= (x + i\sqrt{1 - x^2})^n
 \end{aligned}$$

ii. **2** Hence, prove that:

$$T_n(x) = \frac{1}{2} \left[(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n \right]$$

$$\begin{aligned}
 T_n(x) - i \cdot U_n(x) &= \cos(-n \cos^{-1} x) + i \sin(-n \cos^{-1} x) && \text{sine is odd and cosine is even} \\
 &= (\cos(-\cos^{-1} x) + i \sin(-\cos^{-1}(x)))^n && \text{De Moivre's Theorem} \\
 &= (x - i\sqrt{1 - x^2})^n
 \end{aligned}$$

Adding the two equations, dividing by 2 and putting i inside the square root gives the result.

iii. **3** Consider the series $S = \sum_{n=0}^{\infty} t^n (\cos \theta + i \sin \theta)^n$, where $-1 < t < 1$ and

$$\theta = \cos^{-1} x.$$

Prove that:

$$\sum_{n=0}^{\infty} T_n(x) t^n = \frac{1 - tx}{1 - 2tx + t^2}$$

By the limiting sum of a G.P we have

$$\begin{aligned}
 S &= \frac{1}{1 - t \operatorname{cis} \theta} \\
 &= \frac{1 - t \operatorname{cis}(-\theta)}{1 - t(\operatorname{cis} \theta + \operatorname{cis}(-\theta)) + t^2} \\
 &= \frac{1 - t \cos \theta + i \sin \theta}{1 - 2t \cos \theta + t^2} \\
 &= \frac{1 - tx + i\sqrt{1 - x^2}}{1 - 2tx + t^2}
 \end{aligned}$$

Equating real parts and using demoiivres gives

$$\begin{aligned}
 \sum_{n=0}^{\infty} t^n \cos n\theta &= \frac{1 - tx}{1 - 2tx + t^2} \\
 \sum_{n=0}^{\infty} T_n(x) t^n &= \frac{1 - tx}{1 - 2tx + t^2}
 \end{aligned}$$

9. Question 9 (15 marks)

Start a NEW writing booklet.

- (a) At lunchtime, 5 senior students and 6 junior students decide to line up at the canteen to purchase Chicken Pides. The canteen has three lines for students to purchase food; 2 (fixed) lines for juniors and 1 for seniors. The seniors, having little respect to the juniors, decide that they will line up wherever they want. The juniors, however, stick to their own lines.

- i. **1** In how many ways can the 5 seniors be distributed amongst the 3 lines?

Using spaces and dividers method we have $\frac{7!}{2! \cdot 5!} = 21$ ways

- ii. **3** In how many ways can the 5 seniors and 6 juniors be distributed among the 3 lines?

Ways to distribute the juniors = $\frac{7!}{1! \cdot 6!} = 7$ using spaces and dividers. Multiplying these two together we get 147 ways.

- (b) Consider the general polynomial cubic equation $P(x) = ax^3 + bx^2 + cx + d = 0$ for $a \neq 0$ and a, b, c, d .

$P(x)$ has roots $\alpha, \beta, \gamma \in \mathbb{R}$.

- i. **2** Show that the following polynomial has roots $\alpha + \frac{b}{3a}, \beta + \frac{b}{3a}, \gamma + \frac{b}{3a}$

$$ay^3 + \left(c - \frac{b^2}{3a}\right)y + \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a}\right) = 0 \quad (1)$$

Let $y = x + \frac{b}{3a}$. Subbing this into (1) gives:

$$\begin{aligned}
 a \left(x + \frac{b}{3a}\right)^3 + \left(c - \frac{b^2}{3a}\right) \left(x + \frac{b}{3a}\right) + \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a}\right) &= 0 \\
 a \left(x^3 + \frac{bx^2}{a} + \frac{xb^2}{3a^2} + \frac{b^3}{27a^3}\right) + \left(cx + \frac{bc}{3a} - \frac{b^2x}{3a} - \frac{3b^3}{27a^2}\right) + \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a}\right) &= 0
 \end{aligned}$$

which simplifies to $ax^3 + bx^2 + cx + d = 0$

- ii. **[1]** Let $p = \frac{1}{a} \left(c - \frac{b^2}{3a} \right)$, $q = \frac{1}{a} \left(d + \frac{2b^3}{27a^2} - \frac{bc}{3a} \right)$. Using the substitution $y = s + t$, where s, t are real variables, show that equation (1) is equivalent to equation (2).

$$s^3 + t^3 + (3st + p)(s + t) + q = 0 \quad (2)$$

Dividing both sides of (1) by a we have,

$$y^3 + py + q = 0$$

Substituting $y = s + t$, we have

$$s^3 + t^3 + 3st(s + t) + p(s + t) + q = 0$$

$$s^3 + t^3 + (3st + p)(s + t) + q = 0$$

- iii. **[1]** Suppose that $3st + p = 0$. Show that

$$s^3 + t^3 = -q$$

$$s^3 t^3 = -\frac{p^3}{27}$$

$$s^3 + t^3 + 0 + q = 0$$

$$s^3 + t^3 = -q$$

$$3st + p = 0$$

$$st = -\frac{p}{3}$$

$$s^3 t^3 = -\frac{p^3}{27}$$

- iv. **[2]** Assume without loss of generality that $s^3 > t^3$. Prove that:

$$s^3 = -\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}, \quad t^3 = -\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}$$

Form a quadratic using the expressions in iii with roots s^3 and t^3 . We have:

$$x^2 + q - \frac{p^3}{27} = 0$$

$$x = \frac{-q \pm \sqrt{q^2 + \frac{4p^3}{27}}}{2}$$

$$x = -\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}$$

Now if $s^3 > t^3$, take s^3 as the solution with the positive root and the result is proved.

v. **[2]** Show that the possible solutions of s are:

$$s = \omega^k \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } k = 0, 1, 2$$

where ω is the complex cube root of unity with positive imaginary part.

Let $s = Sz$, where $S = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$.

$$\begin{aligned} S^3 z^3 &= -\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}} \\ z^3 &= 1 \end{aligned}$$

But the solutions to $z^3 = 1$ are of course the cubic roots of unity $1, \omega$ and $\bar{\omega} = \omega^{-1} = \omega^3 \omega^{-1} = \omega^2$ since solutions to a real polynomial occur in conjugate pairs and the conjugate of complex number with modulus one is equal to its inverse. Hence we have $s = S\omega^k$ for $k = 0, 1, 2$.

vi. **[3]** Show that the roots of equation (1) are

$$x = \frac{b}{3a} + \omega^k \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } k = 0, 1, 2$$

where ω is the complex cube root of unity with positive imaginary part.

Similarly to part v), we have:

$$t = \omega^j \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } j = 0, 1, 2$$

Recall the condition $3st = -p$. Letting $T = \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$, we have:

$$3\omega^{j+k}ST = -p$$

Now;

$$\begin{aligned} ST &= \sqrt[3]{\left(-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}\right) \left(-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}\right)} \\ &= \sqrt[3]{\frac{q^2}{4} - \frac{q^2}{4} - \frac{p^3}{27}} \\ &= -\frac{p}{3} \end{aligned}$$

Since p is real, ST must also be real. Hence for $3\omega^{j+k}ST = -p$ to be true, we require both sides to be real i.e ω^{j+k} must be real. One way to ensure this condition is to set $j+k = 3k$ since $\omega^{3k} = (\omega^3)^k = 1$ for $k = 0, 1, 2$. i.e $j = 2k$. Hence by part ii), the roots of (1) are given by $y = \omega^k S + \omega^{2k} T$. But by part i, $y = x + \frac{b}{3a}$. Hence we have

$$x = -\frac{b}{3a} + \omega^k \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad \text{for } k = 0, 1, 2$$